MODEL LAB · 2026-10-16

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)	11	26,271	3.07%	0.6059
served prior	priors compared	held-out plate appearances	held-out HR rate	best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-04, scored on the 26,271 that came after (actual HR rate 3.07%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5794	1.65×	0.02968	0.13619	0.00571
KNN comps Prior is the pooled rate of the nearest comparables.	0.6059	1.56×	0.02962	0.13488	0.00390
SVR Support-vector regression on the same feature matrix.	0.5972	1.58×	0.02967	0.13558	0.00596
Random forest Forest regression; a learned similarity metric.	0.5982	1.53×	0.02964	0.13516	0.00523
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6059	1.56×	0.02962	0.13488	0.00390
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6059	1.56×	0.02962	0.13488	0.00390
Linear regression Ordinary least squares on the same features.	0.6017	1.58×	0.02965	0.13518	0.00398
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6027	1.65×	0.02966	0.13524	0.00623
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6007	1.60×	0.02964	0.13512	0.00470
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5988	1.58×	0.02966	0.13527	0.00582

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **251** knn **251** svr **251** rf **251** core **251** ensemble **251** linear **251** logistic **251** nn **251** xgb **251**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (2,786 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5860	1.51×	0.02314	0.11229	0.01037
KNN comps Prior is the pooled rate of the nearest comparables.	0.6232	1.97×	0.02306	0.10992	0.00977
SVR Support-vector regression on the same feature matrix.	0.6017	1.82×	0.02325	0.11321	0.01716
Random forest Forest regression; a learned similarity metric.	0.6106	2.12×	0.02316	0.11068	0.01542
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6232	1.97×	0.02306	0.10992	0.00977
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6232	1.97×	0.02306	0.10992	0.00977
Linear regression Ordinary least squares on the same features.	0.6211	1.82×	0.02326	0.11135	0.01418
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6307	1.66×	0.02325	0.11095	0.01676
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6236	1.82×	0.02308	0.11000	0.01268
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6196	1.97×	0.02326	0.11125	0.01972

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jonathan Aranda 13.2%, Jo Adell 12.7%, Junior Caminero 12.5%, Patrick Bailey 12.0%, Chase DeLauter 11.0%, Victor Mesa Jr. 10.6%
Flat league	5/6	Jonathan Aranda 12.5%, Jo Adell 12.0%, Junior Caminero 12.0%, Patrick Bailey 11.4%, Chase DeLauter 11.1%, Travis Bazzana 10.5%
KNN comps	6/6	Jonathan Aranda 13.3%, Jo Adell 12.4%, Junior Caminero 12.4%, Patrick Bailey 12.1%, Chase DeLauter 11.1%, Victor Mesa Jr. 10.6%
SVR	6/6	Junior Caminero 13.5%, Jo Adell 13.0%, Jonathan Aranda 12.9%, Patrick Bailey 11.9%, Victor Mesa Jr. 11.2%, Chase DeLauter 10.8%
Random forest	6/6	Jonathan Aranda 13.2%, Junior Caminero 12.8%, Jo Adell 12.6%, Patrick Bailey 11.6%, Victor Mesa Jr. 11.1%, Chase DeLauter 10.9%
Core (KNN+SVR)	6/6	Jonathan Aranda 13.3%, Jo Adell 12.4%, Junior Caminero 12.4%, Patrick Bailey 12.1%, Chase DeLauter 11.1%, Victor Mesa Jr. 10.6%
Forest blend	6/6	Jonathan Aranda 13.3%, Jo Adell 12.4%, Junior Caminero 12.4%, Patrick Bailey 12.1%, Chase DeLauter 11.1%, Victor Mesa Jr. 10.6%
Linear regression	6/6	Jonathan Aranda 13.2%, Jo Adell 12.7%, Junior Caminero 12.5%, Patrick Bailey 11.8%, Victor Mesa Jr. 10.9%, Chase DeLauter 10.8%
Logistic regression	6/6	Jo Adell 13.0%, Jonathan Aranda 13.0%, Junior Caminero 12.9%, Patrick Bailey 11.5%, Victor Mesa Jr. 10.8%, Chase DeLauter 10.7%
Neural net (torch)	6/6	Jonathan Aranda 13.0%, Junior Caminero 12.9%, Jo Adell 12.9%, Patrick Bailey 11.7%, Victor Mesa Jr. 11.0%, Chase DeLauter 10.7%
XGBoost	6/6	Junior Caminero 13.4%, Jonathan Aranda 13.1%, Jo Adell 12.8%, Patrick Bailey 11.5%, Victor Mesa Jr. 10.8%, Chase DeLauter 10.7%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jonathan Aranda**	TB slate	**13.2%**	12.5%	13.3%	12.9%	13.2%	13.3%	13.3%	13.2%	13.0%	
2	**Jo Adell**	CLE slate	**12.7%**	12.0%	12.4%	13.0%	12.6%	12.4%	12.4%	12.7%	13.0%	
3	**Junior Caminero**	TB	**12.5%**	12.0%	12.4%	13.5%	12.8%	12.4%	12.4%	12.5%	12.9%	
4	**Patrick Bailey**	CLE	**12.0%**	11.4%	12.1%	11.9%	11.6%	12.1%	12.1%	11.8%	11.5%	
5	**Chase DeLauter**	CLE	**11.0%**	11.1%	11.1%	10.8%	10.9%	11.1%	11.1%	10.8%	10.7%	
6	**Victor Mesa Jr.**	TB	**10.6%**	10.4%	10.6%	11.2%	11.1%	10.6%	10.6%	10.9%	10.8%	
7	**Travis Bazzana**	CLE	**10.5%**	10.5%	10.4%	10.2%	10.3%	10.4%	10.4%	10.4%	10.1%	
8	**Nathaniel Lowe**	CLE	**10.4%**	10.2%	10.5%	10.4%	10.4%	10.5%	10.5%	10.4%	10.4%	
9	**Yandy Díaz**	TB	**10.0%**	10.3%	10.0%	10.5%	10.2%	10.0%	10.0%	10.1%	10.0%	
10	**José Ramírez**	CLE	**9.4%**	9.7%	9.5%	9.6%	9.6%	9.5%	9.5%	9.8%	10.1%	
11	**Petey Halpin**	CLE	**8.6%**	9.5%	8.6%	9.0%	8.8%	8.6%	8.6%	9.7%	9.2%	
12	**Brayan Rocchio**	CLE	**6.6%**	7.1%	6.5%	6.5%	6.1%	6.5%	6.5%	6.3%	6.3%	
13	**Liam Hicks**	TB	**6.4%**	7.3%	6.4%	7.2%	6.3%	6.4%	6.4%	6.7%	6.6%	
14	**Jonny DeLuca**	TB	**5.9%**	6.4%	5.7%	5.8%	5.5%	5.7%	5.7%	5.7%	5.9%	
15	**Richie Palacios**	TB	**5.7%**	6.5%	5.8%	5.5%	5.2%	5.8%	5.8%	5.6%	5.6%	
16	**Steven Kwan**	CLE	**4.6%**	7.7%	5.0%	3.2%	3.7%	5.0%	5.0%	3.2%	4.7%	
17	**Taylor Walls**	TB	**3.8%**	5.3%	3.9%	3.2%	3.4%	3.9%	3.9%	4.0%	3.9%	
18	**Chandler Simpson**	TB	**2.6%**	5.7%	3.3%	1.9%	1.5%	3.3%	3.3%	1.4%	3.0%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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